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Partisan politics, welfare states, and environmental policy outputs in the OECD countries, 1975–2005

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Abstract

Building on the burgeoning literature on the association between the welfare state and the environmental state, this study empirically examines how the politics of the former has affected the development of the latter. We suggest that the size of the welfare state shapes the calculus of environmental policy costs by partisan governments. A generous welfare state lowers the costs perceived by the left-wing government, as large redistributive spending allows the government to mitigate the adverse impact of the new environmental policy on its core supporters, industrial workers. A generous welfare state also implies diminished marginal political returns from additional welfare commitment by the left-wing government, which lowers the opportunity costs of environmental policy expansion. To the contrary, because of lower overall regulatory and taxation pressure, a small welfare state reduces the costs of environmental policy expansion as perceived by a right-wing government. Our theoretical narrative is supported in a dynamic panel data analysis of environmental policy outputs in 25 Organisation for Economic Co-operation and Development member states during the period 1975–2005.

Keywords: environmental regulation, government partisanship, welfare state.

1. Introduction

Since the late 1960s, states have set up comprehensive administrative structures and implemented a broad range of policies for the purpose of combatting environmental problems. This expansion of state responsibility in the environmental area has led some scholars to suggest the emergence of an environmental state (Barry & Eckersley 2005; Meadowcroft 2005; Duit 2014). The environmental state is often viewed as analogous to the welfare state in that both use regulatory powers to address market externalities (Christoff 2005; Jahn 2014). Just as welfare state expansion entails increasing governmental intervention in the market for the purpose of providing social goods, environmental policy expansion essentially means increasing market intervention by governments in providing otherwise under-provided environmental goods.

The analogy between the welfare state and the environmental state has encouraged scholars in comparative environmental politics to utilize the analytical perspectives advanced in the welfare state literature (Gough & Meadowcroft 2011; Meadowcroft 2012). A long-standing focus of the welfare state literature on class-based distributive conflicts and institutional configurations of political representation has influenced researchers in comparative environmental politics. A fair number of empirical studies have examined how government partisanship, corporatist institutions, and electoral systems affect a country's environmental performance (Fiorino 2011; Cao *et al.* 2014). More recently, studies have also examined how the size of the welfare state is associated with national environmental performance (Jahn 2014; Koch & Fritz 2014). As we discuss in Section II, however, the findings of these studies are largely inconclusive.

In this paper, we seek to move beyond the analogy between the welfare state and the environmental state, and instead explore the dynamic interaction between the two. For this purpose, we develop a theoretical framework specifying how partisan politics in welfare states translate demands for environmental protection into regulatory outputs.

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We begin by establishing that environmental demands do not square well with representative democracy in general and class-based political cleavages in particular. While governments face an array of environmental policy demands from domestic and international sources, the electoral rewards from such policies are confined to relatively small sections of the electorate or to diffused legitimation gains (i.e. having a green record). Our focus is then placed on examining how the largely bipartisan environmental policy demands are filtered through a decisionmaking process still structured heavily by traditional partisan politics. We suggest that when confronted with demands for environmental protection, office-seeking governments across the left–right ideological spectrum strive for minimizing policy costs associated with adopting a new environmental policy. There are two types of costs involved. First, there are direct costs of the environmental policy on core constituencies, for example, workers' job losses for leftist governments and increased production costs or administrative burdens for businesses on the part of rightist governments. Second, there are opportunity costs. Pursuing environmental policy expansion requires political as well as budgetary resources that could have been used for the government's primary policy platforms: redistributive policies for leftist governments and pro-business policies for rightist governments. The lower these two types of costs are, the more likely governments are to adopt a new environmental policy.

This is where the welfare state comes into our theoretical narrative. The size of the welfare state affects the cost calculus of environmental policy adoption by partisan governments. More specifically, a generous welfare state reduces the costs of environmental policy expansion perceived by a left-wing government. A generous welfare state not only mitigates the direct costs of environmental policies accrued to workers, but also implies that the new environmental policy's opportunity cost (i.e. not being able to engage further in redistributive policies) is low. On the contrary, a small welfare state reduces the costs of environmental policy expansion for a right-wing government. A small welfare state offers businesses greater flexibility to maneuver when a new environmental regulation puts upward pressure on production costs. A small welfare state also implies that the opportunity costs of the new environmental policy (i.e. not being able to expand pro-business policies) are low. In essence, a left (right)-wing government is more likely to expand environmental policy in times when the welfare spending is relatively large (small). Our dynamic panel data analysis of environmental policy outputs in 25 Organisation for Economic Co-operation and Development (OECD) member states over a 30-year period (1975–2005) lends support to our theoretical framework.

2. Literature and contribution

In this section, we review existing studies on government environmental performance that focus on our two key variables of interest: government partisanship and the welfare state. We then discuss how this paper contributes to the existing literature both theoretically and methodologically.

2.1. Does partisan politics explain government environmental performance?

A significant number of studies have explored whether traditional partisan politics can explain government performance in protecting the environment. Taking an approach equivalent to class-analytical accounts in the welfare state literature (Korpi 1980, 1989; Huber *et al.* 1993), some suggest that a left-wing government is more pro-environmental than a right-wing government as the former identifies more with the victims of environmental problems. For instance, the urban poor, who often identify themselves with the left party, are more likely to be victims of industrial pollution and are therefore also more likely to benefit from the expansion of urban environmental protection. Accordingly, successful political mobilization of environmental demands has been part of a wider left-wing agenda (Pepper 1984). Right-wing parties, to the contrary, are often associated with a less ambitious green agenda because environmental policies impose costs on businesses (Dunlap & Gale 1974; Dunlap & Allen 1976).

This, however, might well be an oversimplification. The traditionally more organized supporters of the left party, labor unions, have reasons to oppose certain measures for environmental protection.[1] Already in the early 1980s, increased environmental regulations in the United States' fossil-fueled electric power industry were find to have led to significant job loss in the sector (Gollop & Roberts 1983). Some suggest that "Unionized blue-collar workforces, particularly those in manufacturing and extractive industries" are more likely to face "job loss and economic uncertainty if environmental policy threatens the profitability of their industry" (Obach 2002, p. 83).

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What is perhaps more important in the context of electoral politics than the actual impact of an environmental policy on jobs, is the *perceived* impact on core constituents. The so-called "jobs versus the environment" sentiment may shape the negative attitudes of left party voters toward environmental regulation, even when the overall/average relationship between environmental regulation and jobs is positive (Morgenstern *et al.* 2002). Isolated yet salient industry or firm-level experience (e.g. the fossil-fueled electric power industry) can drive such negative attitudes. As early as 1990, a poll conducted by the *Wall Street Journal* finds that one-third of the respondents thought it "somewhat or very likely that their own job was threatened by environmental regulation" (Rosewicz 1990). Also, if the job creation effects of environmental regulations are more concentrated on some industries (manufacturing and professional, information, scientific, and technical services according to Bezdek *et al.* 2008), other industries that do not benefit as much may sustain the "jobs versus the environment" attitude. Indeed, recent studies find that there is a significant sectoral or industry-wise variation in environmental demands among the supporters of the left party. For example, a recent survey analysis conducted in Sweden shows that unions in production industries tend to take a much more reluctant stance toward progressive climate policies than unions in other sectors (Gingrich 2013).

Likewise, the traditionally more organized supporters of the right party, businesses, might not necessarily oppose stringent environmental regulations. Certain environmental policies may have little undermining effect on business profitability or even enhance production efficiency. A case in point is the US waste processing industry that adopted a labor-saving, automatized recycling system to the detriment of employees (Weinberg *et al.* 2000). Businesses might also demand stringent environmental policies requiring specific technologies that allow local businesses to enjoy the first mover, or the top runner, advantage (Vogel 1997). Japan had a strong "eco-modernizing coalition" during the 1970s thanks to the "environmental optimism within parts of the industry, and in fields such as renewable energy and energy efficiency," which encouraged its right-wing government to adopt the hard-core pioneer role in environmental policy amounting to "green protectionism" (Jänicke 2005, pp. 132, 136). More recently, the right-wing government in British Columbia, Canada surprised many observers by adopting a comprehensive carbon tax scheme (Harrison 2010). Essential in this process was the absence of any backlash from businesses. Business communities were indeed willing to grant a greater leeway to the government led by the self-proclaimed "free enterprise" Liberal Party than it would have been to the left-wing government led by the social-democratic New Democratic Party (Harrison 2013).

Findings from a few large-N empirical studies mirror this Janus-faced relationship between business, labor, and environmental regulation. Some find a positive association between left-libertarian party strength and national environmental performance measured using the pollution index (Jahn 1998). Others understand this relationship to be more nuanced. Neumayer (2003), for instance, finds that a strong presence of green/libertarian left-wing parties is associated with lower emissions of major air pollutants, but he does not find the effect of traditional left-wing parties to be consistent over different models in his study. Still others do not find any significant association between the traditional left–right partisanship and environmental performance measured by policy outputs (Knill *et al.* 2010).

2.2. Are generous welfare states more committed to environmental protection?

What about the effect of the welfare state on environmental performance? If the rationale for expecting diverging environmental policy preferences has in part to do with the preferred "scope of government activities, especially in an effort to solve perceived societal problems" (Dunlap & Allen 1976, p. 395), one should expect a large welfare state to be more pro-environmental overall than a small welfare state. After all, both environmental and welfare policies are about using state powers to address market externalities (Christoff 2005; Jahn 2014; Koch & Fritz 2014). Experience with the interventionist welfare institutions might allow citizens to be more favorable to the interventionist environmental policy, regardless of their partisan allegiances.

In this vein, scholars have argued that (neo)corporatist institutions in governing labor-business relations have a positive effect on environmental protection (Jahn 1998; Scruggs 1999, 2001). The institutional scope and centralization arguably allow both labor and employers to better consider the negative environmental externalities of their decisions and pursue joint solutions, thus diffusing "best practices" more readily throughout the economy. Yet some small-N case studies of corporatist countries find that corporatism does not always facilitate environmental policymaking. Mol's (2003, p. 342) comparative study of four European countries suggests that the traditional corporatist institutions "do not succeed in maintaining their dominant role" in joint environmental policy arrangements. The study furthermore suggests that the effect of corporatist institutions may vary over time and/or by policy types.

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Some recent studies have moved on to examine the direct effect of the welfare state on national environmental performances. Kerret and Shvartzvald's (2012) cross-country study finds a positive association between the social policy index (SPI) and the environmental performance index (EPI). Bernauer and Böhmelt's (2013) cross-sectional study, however, finds no robust evidence for the claim that countries with better social policies are also likely to be greener, when outliers are removed and the EPI index decomposed. Koch and Fritz (2014) investigate whether country clusters along welfare state typologies (i.e. liberal, conservative, and social democratic welfare states) correspond with country clusters based on sustainability indicators or what they call "eco-state clusters." They find little evidence for correspondence. Instead, countries from all welfare regimes are "spread across established, deadlocked, failing, emerging and endangered eco-states" (Koch & Fritz 2014, p. 679).

2.3. Why are existing studies inconclusive?

Why do we encounter such inconclusiveness in the existing literature? One quick answer may be a lack of consistency between theoretical conceptualization and empirical operationalization of environmental performance. The vast majority of the existing empirical literature focuses on environmental outcomes, such as selected emission statistics or a composite index of multiple outcome indicators, such as the EPI (Emerson *et al.* 2012). This is the case even when the causal story being put to test focuses explicitly on the scope of governmental engagement with environmental problems, or, in other words, policy outputs. Relying on outcome indicators inevitably leads to black-boxing the better part of the environmental policy dynamic, which, in turn, precludes the understanding of how environmental problems are transformed into environmental policy responses (Meadowcroft 2014). Environmental policy output data so far have been limited in both geographical and temporal coverage (see, however, Knill *et al.* 2010, 2011 for exceptions). As a partial remedy to this problem, this paper utilizes a newly extended dataset on national environmental policy outputs that encompasses 25 broad environmental policy areas. Our empirical focus on policy output is consistent with our theoretical focus on the government as the main actor who decides whether to engage in environmental policy expansion or not.

A second and related limitation of previous work is that most rely on static, cross-sectional comparisons. We suggest that environmental policy expansion should be viewed as a process that unfolds over long time periods, and that the empirical analysis should take the nature of the process into account. As was the case in the welfare state development, the process of the environmental state development is expected to entail alternating periods of innovation, relative stagnation, and consolidation (Meadowcroft 2005). The effects partisan politics and the welfare state have on environmental policy expansion might not be immediate but instead distributed over future time periods. Our dynamic panel data analysis covering the period 1975–2005 enables us to move beyond the limitations of snapshot between-country comparisons.

Finally, existing studies tend to draw on the analytical parallels between welfare and environmental policies. Only recently have some conceptual works on the environmental state started to acknowledge the difference between the two policy areas and the potential interaction between them. For instance, Gough and Meadowcroft (2011) note that in most advanced democracies, welfare policies have traditionally "trumped" environmental policies. The reasons are twofold. First, the socioeconomic risks that the welfare policies seek to address "bite harder and/or more rapidly than do indirect effects of environmental deterioration" (Gough & Meadowcroft 2011, p. 495) Second, welfare systems have over time "nurtured interest coalitions" within the traditional political scene. Predicting a government's move toward or against environmental regulation thus requires one to understand first the government's commitment to respond to socioeconomic risks, and then to assess how the new environmental policy may affect such commitment.

3. Theoretical framework

The main actor in our theoretical framework is a representative government that seeks political survival. We assume that ruling political parties (coalitions) across the left–right ideological spectrum have distinct policy commitments in areas where they can strongly appeal to their core constituencies. These are usually policy areas that involve tangible (re)distribution of resources. A government led by a left-wing party/coalition would want to expand redistributive spending and maintain it at a high level to offer distributive benefits to workers, whereas a government led by a right-wing party/coalition would want to reduce such spending or maintain it at a low level to protect business interests. Such commitments are predominantly, although not exclusively, reflected in social policy. Across countries and over time, left-wing governments have sought to increase the size of the welfare state, while right-wing governments have sought its retrenchment (Korpi 1980, 1989; Huber *et al.* 1993; Garrett 1998; Allan & Scruggs 2004).

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Unlike social policy, environmental policy squares poorly with traditional partisan politics. While environmental issues are gaining salience among citizens in most industrialized countries (Dunlap & York 2012), studies have shown that core supporters of environmental policy expansion are "clearly different both from manual workers as well as from the old middle class, the two antagonists of the traditional economic division" (Dolezal 2010, p. 537). As highly educated, young, libertarian urban voters tend to lend support for Green parties, the mainstream parties across the political spectrum have been incentivized to form electoral alliance with the Green parties or to incorporate environmental agendas in their own platforms (Dryzek *et al.* 2002; List & Sturm 2004; O'Neill 2012). Organized domestic environmental lobby groups actively, and sometimes successfully, call for greater government intervention in environmental protection (Fredriksson *et al.* 2005). Some suggest that environmental policy has gradually become one of the key "valence" issues in electoral politics as a result of the increasing problem pressures and issue awareness (Rohrschneider 1988; Hofrichter & Reif 1990; Green 2007).

There have also been international pressures for environmental policy expansion. Upward regulatory competition for sustaining or catching up with lead markets in green technology is found to be an important driver of national environmental policy expansion in environmental pioneer countries (Vogel 1997; Jänicke 2005). For the newly industrialized countries in Eastern Europe, Latin America and Asia, joining the OECD club of rich countries or the European Union (EU) has been an important driver of environmental policy expansion (Knill *et al.* 2008; Sommerer & Lim 2016). The remarkable environmental policy expansion observed in the OECD economies over the last three decades can be viewed as the joint effect of all of these demands (Fiorino 2011; Weidner & Jänicke 2002; Duit 2014).

So what role does traditional partisan politics play in this process? We suggest that even when there are strong demands for environmental regulations that parties across the ideological spectrum cannot ignore, the process of translating such demands into a concrete policy output still involves partisan politics to its core. This is because governments across the left–right ideological spectrum have distinct and persistent commitments to their primary policy platforms that appeal to their core supporters. Office-seeking governments are unwilling to compromise or jeopardize these platforms for the sake of producing environmental regulations. In essence, at a given level of environmental demand, environmental policy expansion is costlier and therefore less likely when it is perceived to undermine the government's primary policy platforms.

This is how welfare state politics comes into our theoretical narrative. The size of the existing welfare state affects the cost calculus of a partisan government faced with environmental demands. More specifically, the welfare state shapes the calculus of two distinct types of costs. First, there are direct costs of the environmental policy accrued to the government's core supporters, for example, workers' job losses for a left-wing government and increased production costs or administrative burdens for businesses on the part of a right-wing government. These costs directly undermine the government's distributive commitments. In any given country, a left-wing government would evaluate whether a new environmental policy may have a negative (or perceivably negative) effect on wage and employment prospects. A left-wing government is more likely to engage in environmental policy expansion when such (perceived) risks on workers well-being can be contained or when it can convince workers so. Because redistributive social spending can mitigate the environmental policy's adverse effect on workers, such as job losses, a left-wing government is more likely to expand environmental policy under a generous welfare state. On the other hand, a right-wing government would fear that a new environmental policy might affect production costs and profit margin of businesses. It is more likely to engage in environmental policy expansion when such costs on businesses are expected to be low or when it can convince businesses that the costs can be contained. Thus, the right-wing government is more likely to adopt a new environmental policy under a small welfare state, which gives more flexibility for the businesses to recover regulatory costs.

Second, there are opportunity costs. Environmental policy expansion involves spending political resources that could have been used for the government's primary policy platforms directed at core constituencies, for example, redistributive social policies for a left-wing government and pro-business policies for a right-wing government. When the opportunity costs are high, the government is unlikely to engage in environmental policy expansion. Yet as the marginal returns from primary policy platforms decrease, the opportunity costs involved in environmental policy expansion also decrease. This may spur a government to pass environmental policies in order to expand its support base into sections of the electorate for which green policies are a primary policy preference, and to strengthen their overall green image in the eyes of the electorate as a whole. In any given country, a left-wing government would be more willing to adopt a new environmental policy under a generous welfare state. This is because when there is already a generous welfare state, the marginal returns to the government of further expanding redistributive social spending is small. To the contrary, a right-wing government would be more willing to pursue environmental policy expansion under a small

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welfare state, as the marginal returns from additional pro-business policies are small when the size of redistributive spending is already kept at low levels.

Our main hypothesis is as follows:

Hypothesis 1: *The effect of government partisanship on environmental policy expansion is conditioned by the size of the welfare state.*

More specifically,

Hypothesis 1.1: *A generous welfare state facilitates environmental policy expansion by a left-wing government.*

Hypothesis 1.2: *A small welfare state facilitates environmental policy expansion by a right-wing government.*

Hypothesis 1 can be pitted against two alternative hypotheses suggested by the existing studies we surveyed earlier. One draws from the literature on the partisan divide in environmental politics, especially in the context of American politics (Dunlap & Gale 1974; Dunlap & Allen 1976; Guber 2001, 2013):

Hypothesis 2: *A left-wing government is associated with greater environmental policy expansion than a right-wing government (regardless of the size of the welfare state).*

Another draws from the literature on the functional analogy, and thereafter synergy, between the welfare state and the environmental state, developed in the context of European welfare states (Christoff 2005; Jahn 2014; Koch & Fritz 2014):

Table 1 Policy items included in environmental policy scope

Policy	Environmental Issue Area	
Sulphur content, gas, oil	Pollution	Air
Lead content in petrol	Pollution	Air
Passenger car emissions	Pollution	Air
Large combustion plant emissions	Pollution	Air
CO2 tax or trading	Pollution	Air
DDT ban	Pollution	Air/Water/Soil
Industrial discharge, surface water	Pollution	Water
Hazardous substances in detergents	Pollution	Water
Coliforms in bathing water	Pollution	Water
Noise emission standard for lorries	Pollution	Noise
Motorway noise emissions	Pollution	Noise
Contaminated site clean up	Pollution	Soil
Waste landfill target	Pollution/Resource Depletion	
Soil protection	Resource Depletion	
Feed in tariffs	Resource Depletion/Pollution	
Efficient use of water in industry	Resource Depletion	
Energy efficiency of refrigerators	Resource Depletion	
Recycling construction waste	Resource Depletion	
Glass reuse recycling target	Resource Depletion	
Paper reuse recycling target	Resource Depletion	
Endangered species protection	Resource Depletion/General	
Environmental impact assessment	General	
Eco labeling	General	
Eco audit	General	
Environmental sustainability plan	General	

DDT, dichloro-diphenyl-trichloroethane.

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Hypothesis 3: *A generous welfare state facilitates environmental policy expansion (regardless of government partisanship).*

4. Empirical Analysis

Our hypotheses are tested in dynamic panel data analysis of 25 environmental policy outputs in an unbalanced panel of 25 OECD countries during the period 1975–2005.[2] Our sample includes: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Japan, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, South Korea, Spain, Sweden, Switzerland, the United Kingdom and the US.[3]

4.1. Dependent variable

Our dependent variable is environmental policy output. We collected data on the enactment of 25 different environmental policies, which include a broad range of environmental issues and policy instruments (see Table 1).[4] Official governmental records and documents, legal collections, and secondary literature were examined to locate policies. Only those policies issued by the executive or parliament were included in the dataset, and each of the 25 policies was defined by a set of criteria that must be fulfilled for it to be counted as a policy. As the total number of environmental policies in a given country is unknown, it was not possible to select policies based on random sampling. We nevertheless believe that these policies are representative of a larger policy universe and that the policy expansion in these areas reflects governments' overall environmental policy commitment.

As our main dependent variable, we create the *environmental policy scope*. It is a proportional variable ranging between 0 and 1, and takes the value of 1 when a country-year observation has all 25 sample policies in place. While the actual cost of a specific policy is determined by the economic, physical, and political context in which the policy is designed and implemented, our variable is based on the simple assumption that more policies take more commitments than fewer policies. Following a conventional technique, we logit-transform the proportional variable (Aitchison 1986).[5] The transformation addresses the bounded nature of the observed outcome and allows the effect of an independent variable on a proportional dependent variable to be non-linear.[6]

For robustness checks, we employ two alternative operationalizations of the dependent variable: raw number of policies in place (ranging from 0 to 25), and adoption of a policy as a binary outcome variable (0 when the policy is absent in a given area, and 1 when the policy is adopted).[7] As an additional robustness check, we employ an alternative indicator of environmental policy output, environmental tax per capita (in US$100), to see if our framework can also explain the depth of a single environmental policy, as opposed to the mere introduction of a policy.

4.2. Key independent variables

Our key independent variables are government partisanship (left vs. right) and the size of the welfare state. For government partisanship, we use a dummy variable. Government partisanship equals 1 where there is a left-wing governing party (/coalition) and 0 where there is a right-wing governing party (/coalition).[8] For a small number of observations (less than five percent) governed by center parties, we assign the value of 1 as center parties have more often aligned with left-wing than right-wing parties in the countries in our sample. Our results do not change when we drop the observations that are governed by center parties. Importantly, we do not use vote/seat share operationalized in a percentage scale because our theoretical framework centers on partisan dominance in government. While the actual threshold vote/seat share for enjoying a dominant position may vary across countries, a percentage increase in right (left) parties' vote/seat share is unlikely to have any substantive effect on environmental policy outputs without reaching that threshold.

For the size of the welfare state, we use the budgetary social protection outlay as a percentage of total central government outlays. The social protection outlay includes spending on various social insurance programs, means-tested and in-kind social assistance programs, as well as research and development on all of these programs. The measure

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is expected to capture the prioritization of the redistributive goals of the welfare state over many other policy objectives under a given budget constraint.[9]

4.3. Controls

We include a set of variables to control for the variation in external and domestic sources of environmental policy demands. First, we control for three core indicators of economic modernization: gross domestic product per capita (in constant 2007 US dollars and adjusted for purchasing power parity), deindustrialization, and urbanization.[10] These indicators have been argued to have a positive association with environmental protection (Mol *et al.* 2009; O'Neill 2012). Second, we control for trade dependency, EU membership (the year when the accession process begins), and environmental treaties membership, all of which capture externally originated pressures on environmental quality (Holzinger *et al.* 2008; Vogel 1997).[11] Another potential driving force of environmental policy expansion is environmental administration, which most countries began to set up in the 1960s. We control for the presence of the ministry of the environment and/or an environmental protection agency (Weidner & Jänicke 2002).

In addition to the basic set of control variables, we introduce four additional control variables. As adding these variables substantially reduces the sample size, we introduce them in selected models only. First, we add an indicator of political-institutional constraints that might independently affect the ease and incentives for producing environmental public goods. More specifically, we control for the time-varying veto points using Henisz's (2006) Political Constraint Index Dataset (POLCON) III. We further weigh the veto points with the electoral system (proportional = 1, majoritarian = 2), as it has been suggested that the majoritarian system entails stiffer competition than the proportional system, as elected politicians are more direly and separately accountable to the voters (Persson & Tabellini 1999). In a majoritarian system, politicians end up competing more fiercely for transfers while underproviding public goods (such as environmental protection) that benefit a broader segment of the population (Fredriksson *et al.* 2005; Ward & Cao 2012). Thus, our intuition is that institutional constraints would be highest when a government elected by a majoritarian electoral system also faces many time-varying veto points (e.g. by having a divided government). We expect the electoral system-weighted veto points to have a *negative* association with the environmental policy scope.

Second, we add the unemployment rate to capture potential shocks on the labor market. This variable is important in two regards. First, the rate of unemployment as a proxy for economic cycles may directly affect government's policy prioritization.[12] Second, and perhaps more importantly for testing our hypothesis, a short-term fluctuation in labor market conditions (e.g. as a result of business cycles) might result in fluctuations in social spending that have little to do with the partisan welfare state commitment (e.g. as a result of a sudden increase in unemployment benefits applications). Controlling for the unemployment rate thus allows us to tease out the social spending that reflects the longer-term welfare state commitment from that which reflects short-term fluctuations.

We also control for environmental issue saliency among political parties. Increased environmental issue salience creates new opportunities for environmental movements and Green parties. It also encourages traditional parties to develop positions on environmental issues, and at times, to enter into strategic alliances with the Greens. To capture the growing salience of the environmental issue, we calculate the vote-weighed average share of environmental manifestos of all parties that competed in the latest preceding election.[13] The weighing scheme means that our measure of environmental issue salience is correlated with a strong presence of the green party. We prefer the weighted environmental manifesto variable to other indicators specific to the presence of green parties in parliament because the former can capture the over time variation in environmental issue salience, even in countries where there have been no green parties in parliament. We expect the variable to have a positive association with environmental policy scope.

Finally, we control for the level of corporatism by employing a time-varying corporatism index recently made available (Jahn 2016). This is to capture how "corporatism in flux" may affect environmental policy making (Mol 2003, p. 342). The index operationalizes three conceptual aspects of corporatism (organizational-structural, functional, and scope) using the database on Institutional Characteristics of Trade Unions, Wage Setting, State Intervention and Social Pacts (ICTWSS). The richness of the index allows us to control for changes over time in corporatist arrangements within a country, which might independently shape environmental policy output. Extending the insights from the

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existing large-N studies on the corporatism and environmental performance nexus, we expect the corporatism index to have a positive association with environmental policy scope (Jahn 1998; Scruggs 1999, 2001).

4.4. Model

We model our main dependent variable, *environmental policy scope*, using the error correction model (ECM). We chose the ECM over other forms of nested/restricted dynamic specifications for two reasons. The first is to avoid imposing potentially invalid restrictions on the functional form of the lag structure.[14] Another important advantage of the ECM is that "variables are parameterized in terms of changes, helping to avoid spurious findings if the stationarity of the series is in question" (De Boef & Keele 2008, p. 195). While we do not find any indication of non-stationarity

Table 2 Determinants of environmental policy expansion (1975–2005)

	Model 1 (Null)	Model 2	Model 3	Model 4	Model 5
Instantaneous Effect					
Δ_Left government	−0.011	0.017	−0.002	0.033	−0.019
Δ_Welfare spending	−0.006**	−0.006*	−0.006*	−0.005***	−0.005
Δ_Left x Welfare		−0.001	−0.000	−0.001	0.000
Δ_GDP per capita	0.028	0.030	0.081	0.053	−0.064
Δ_GDP per capita^2	−0.001	−0.001	−0.003	−0.002	−0.000
Δ__Urbanization	0.090	0.083	0.164**	0.189*	0.116
Δ_Trade	0.001	0.000	−0.000	0.000	0.000
Δ_Deindustrialization	0.002	0.002	−0.002	0.002	−0.003
Δ_Environmental administration	0.127	0.128	−0.104	−0.075	−0.086
Δ_EU	0.002	0.001	−0.003	−0.006	−0.029
Δ_Environmental treaty	0.411	0.408***	0.705*	0.730$	0.888*
Δ_Unemployment			−0.013	−0.014	−0.021**
Δ_Political constraints			0.076	0.072	0.087
Δ_Environmental party manifesto				−0.005	−0.006
Δ_Corporatism					0.006
Long Run Multiplier Effect					
Left government	−0.269	−1.238***	−1.157*	−1.162****	−1.615****
Welfare spending	−0.045**	−0.062**	−0.039**	−0.040**	−0.046**
Left x Welfare		0.028$	0.033**	0.034****	0.042****
GDP per capita	0.485****	0.522**	0.764**	0.708**	0.605**
GDP per capita^2	−0.011**	−0.012**	−0.014**	−0.013**	−0.012**
Urbanization	0.091**	0.085****	0.085**	0.086*	0.058
Trade	−0.010	−0.012	−0.010	−0.008	−0.001
Deindustrialization	0.066****	0.060$	0.006	0.002	0.022
Environmental administration	0.140	0.141	0.197	0.250	0.147
EU	1.019**	0.997****	0.451	0.515	0.480
Environmental treaty	2.401**	2.508**	1.362****	1.571****	2.359**
Unemployment			0.117****	0.112****	0.127***
Political constraints			−0.037	−0.084	0.114
Environmental party manifesto				0.024	0.045
Corporatism					0.548
Error Correction Rate					
Lagged DV	−0.085**	−0.085**	−0.131**	−0.128**	−0.115**
Observations (Countries)	611 (25)	611 (25)	498 (25)	481 (25)	455 (23)

*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

$$P < 0.1$$

Driscoll and Kraay's variance-covariance matrix estimator is used with the delta method to calculate the standard errors and P values of the long-run multiplier effect. DV, dependent variable; EU, European Union; GDP, gross domestic product.

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in our dependent variable in augmented Dickey–Fuller regressions, political time series often produce conflicting results in tests for stationarity and integration. The ECM is useful in such ambiguous situations (De Boef & Keele 2008). The model equation can be written as follows:

$$\Delta EnvPolicyScope_{it} = \beta_0 EnvPolicyScope_{i,t-1} + \Delta X_{it}\beta_1 + X_{i,t-1}\beta_2 + \alpha_i + \varepsilon_{it} \quad (1)$$

where X is the matrix of covariates, β0 is the coefficient estimate of the lagged dependent variable, and β1 and β2 are vectors of coefficient estimates for first-differenced and lagged values of X, respectively.

Consistent with the literature, we also include country fixed effects: αi, to control for country-specific factors that may influence a country's propensity to adopt environmental policies. The inclusion of country fixed effects can pick up the effects of largely time-invariant political institutions, such as federalism, electoral rules, and legislative organization (e.g. bicameralism) (Fredriksson & Millimet 2004, 2007), term limits for elected officials (Fredriksson *et al.* 2011), and ministerial discretion (Bäck *et al.* 2015).

In a dynamic model, the effect of an independent variable can be examined in two aspects: the instantaneous (or ephemeral) effect and the long-run effect. The changes in the independent variable in some previous time points affect the dependent variable, yet the effect is distributed across several future time periods. For instance, in our case, the level of social spending may affect environmental policy outputs both contemporaneously and into the future. As a result of the nature of policymaking, which often takes several years from proposal to legislation, we expect the theorized effect of partisan politics of the welfare state would be more pronounced in the long run. The ECM specification estimates the instantaneous effect directly, which is captured by β1. To identify the long-run effect an independent variable has on the dependent variable distributed over future time periods, we calculate and report each independent variable's long-run multiplier (LRM) as follows:[15]

$$LRM_x = \frac{\beta_{2x}}{-\beta_0} \quad (2)$$

The delta method is used to calculate the correct standard errors for the LRM effect. The standard errors are adjusted using Driscoll and Kraay's (1998) robust variance–covariance estimator, which is a non-parametric technique designed for panel data estimated with country fixed effects. The estimator addresses both group-wise contemporaneous correlation and heteroskedasticity.

4.5. Findings

The results of our empirical analysis are presented in Table 2. We first fit a null model, where we look at the effect of government partisanship and welfare spending without including the interaction term of the two (Model 1). Both in the short-term and the long run, we find that government partisanship has *no* significant impact on environmental policy expansion. This is against Hypothesis 2, which posited that a left-wing government is associated with greater environmental policy expansion than a right-wing government. Our finding in the null model is consistent with our theoretical underpinning that there is no strong partisan divergence over environmental protection per se. In the null model, we also find that size of social spending has a negative association with environmental policy scope. This finding suggests that rising welfare spending tends to increase the governments' perceived costs of a new environmental policy. This weakens Hypothesis 3, which posited that a generous welfare state facilitates environmental policy expansion (regardless of government partisanship).

In Model 2, we explore how government partisanship and the size of the welfare state interactively affect environmental policy scope. In Models 3–5 we introduce additional control variables: Model 3 adds unemployment rate and political constraint, Model 4 controls in addition for environmental issue salience, and Model 5 further controls for corporatist arrangements. We find a statistically meaningful long-run interaction effect across all interaction models. In light of our main hypotheses (i.e. Hypotheses 1.1 and 1.2 concerning the conditioning role of the size of the welfare state), we interpret the interaction term focusing on the long-run marginal effect of

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government partisanship. Following Brambor *et al.* (2006), we graphically display the long-run marginal effect of left party dominance in government on environmental policy scope at various levels of welfare spending in Figure 1 (based on Model 5). Any particular point on the interior solid sloping line indicates a long-run marginal effect of left party dominance at different levels of welfare spending.[16] The dotted lines are 90 percent confidence intervals.

The long-run marginal effect of left party dominance is *negative* when the welfare spending is below about 35 percent of the budget, suggesting that a left-wing government has an undermining effect on environmental policy expansion when the welfare state is small. Environmental policy scope would have been wider under right party dominance. However, the marginal effect of the left party dominance becomes increasingly *positive* when the welfare budget is above 35 percent of total government outlays. This is when a left party coming into power would facilitate environmental policy expansion. Such an interaction effect is observable across Models 2–5, although the slope of the line and confidence intervals vary slightly. This finding supports Hypotheses 1.1 and 1.2.

The same interaction terms can also be interpreted focusing on the long-run marginal effect of welfare spending, while fixing the government partisanship variable at either the left or the right. In Model 2, for example, we find that an increase in welfare spending undermines environmental policy expansion under both types of partisan dominance, although the undermining effect is stronger under a right-wing government (−0.062) than under a left-wing government (−0.062 + 0.028 = −0.034). In models with additional control variables, the negative effect of increased welfare spending is almost *nullified* for the left-wing government (−0.006 in Models 3–4 and −0.004 in Model 5). The finding reaffirms our theoretical narrative that a left-wing government sees less tension between the welfare state and the environmental state than a right-wing government. In any case, we do not find support for Hypothesis 3, which postulated that a generous welfare state would *facilitate* environmental policy expansion, regardless of government partisanship.

4.6. Robustness checks

We report results from a series of robustness checks in Table 3. Models 6 and 7 tackle concerns about potential casual heterogeneity that might complicate the testing of our theoretical framework. Our theoretical framework is developed in the context of representative democracies in which politicians seek office and are thus responsive to the concerns of their core constituents. While the 25 OECD member countries do represent contemporary representative democracies, several non-Western member countries that are relatively new and less mature democracies are included: Hungary, Mexico, Poland, and South Korea. In Model 6, we exclude them from the analysis to make sure that these four countries did not drive our earlier theory-confirming findings when the applicability of our theoretical narrative is weaker.

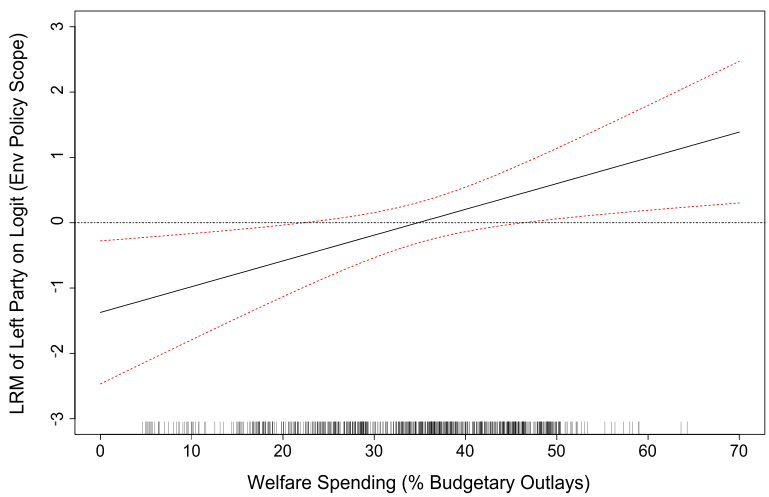

Figure 1 Partisan effect on environmental policy output expansion (based on Model 4). LRM, long-run multiplier.

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Table 3 Robustness Checks

	Model 6	Model 7	Model 8	Model 9	Model 10
	Environmental Policy Scope		No. of Policies (0–25)	Adoption (0/1)	Environmental Tax (US$100 per capita)
DV	LRM			Estimate	LRM
Intercept				−8.193***	
Left government	−1.426*	−1.221***	−5.518**	−0.587	−0.89**
Welfare spending$	−0.050**	−0.046*	−0.128**	−0.028**	−0.02
Left x Welfare$	0.041*	0.029*	0.146**	0.020***	0.03**
GDP per capita$	0.453**	0.043	2.748**	−0.081***	0.44
GDP per capita2§	−0.010**	−0.002	−0.051**	0.001	−0.01
Urbanization	0.087***	0.100*	0.060	−0.001	0.12
Trade$	−0.008	0.002	−0.070***	0.017**	0.00
Deindustrialization$	−0.001	0.064	0.006	−0.002	−0.04
Environmental Administration	0.068	−0.273	0.061	0.502	0.74***
EU	0.495	0.648	2.830*	−0.103	−0.37
Environmental treaty	3.053**	3.372**	9.673**	1.297**	1.70
Unemployment$	0.043	−0.049	0.357***	0.037***	0.09
Political constraints	0.622	0.277	−0.432	0.569	−0.80***
Environmental manifesto	0.048	0.019	0.396**	0.054*	−0.02
Lagged DV	−0.122**	−0.203**	−0.165**		0.34*
N (country)	433 (21)	286 (15)	481 (25)	5298 (25)	210 (23)
	1975–2005				1995–2005

*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

$Variables in Model 9 are included as historic five-year running averages. Model 6 excludes Hungary, Mexico, Poland, South Korea. Model 7 additionally excludes Australia, Austria, Belgium, France, Switzerland, and United States.

Model 10: Second to fifth lag lengths of the dependent variable (DV) are used as generalized method of moments instruments. Sargan Test: chi-square (33) = 23 ($P = 0.90$); AR, autoregressive; (2): normal = -1.20 ($P = 0.23$). EU, European Union; GDP, gross domestic product; LRM, long-run multiplier.

Our main finding with regard to the interaction effect of government partisanship and the size of the welfare state on environmental policy expansion holds.

Another important underpinning of our theoretical framework is the idea that left parties are more committed to the welfare state than right parties. Does this assumption hold across all mature democracies? If our results were driven by the countries in which there is no robust correlation between partisan ideology and welfare state commitment, the finding might indeed be indicative of a spurious, atheoretical correlation. To address this concern, we investigate the party ideology and welfare commitment nexus by utilizing the party manifesto data from the Manifesto Project Dataset (MPD). We calculate, for each country-election year, the seat-weighted "left" orientation of the parliament, as well as the seat-weighted "welfare" orientation of the parliament. The correlated coefficients of the two orientations capture the linkage between the leftness and welfare commitment. Based on the results, we identified six countries in our sample where the linkage was not statistically significant: Australia, Austria, Belgium (P value of the correlation coefficient is 0.06, but we treated this as not significant to remain conservative), France, Switzerland, and the US. In Model 7, we exclude these suspected countries from the analysis. Our main finding on the interaction of government partisanship and the size of the welfare state holds.[17]

We also conduct two technical robustness checks to make sure our finding is not driven by the choice of a specific variable operationalization and/or estimation method. In Model 8, we employ a raw number of policies ranging from 0 to 25, instead of the logit-transformed proportion variable, to show that our main finding is not driven by the operationalization strategy. Our finding on the interactive effect of government partisanship and welfare state on environmental policy output holds. The interaction effect is visualized in Figure 2; the long-run marginal effect on the y-axis can readily be interpreted in terms of the difference in number of policy outputs. In

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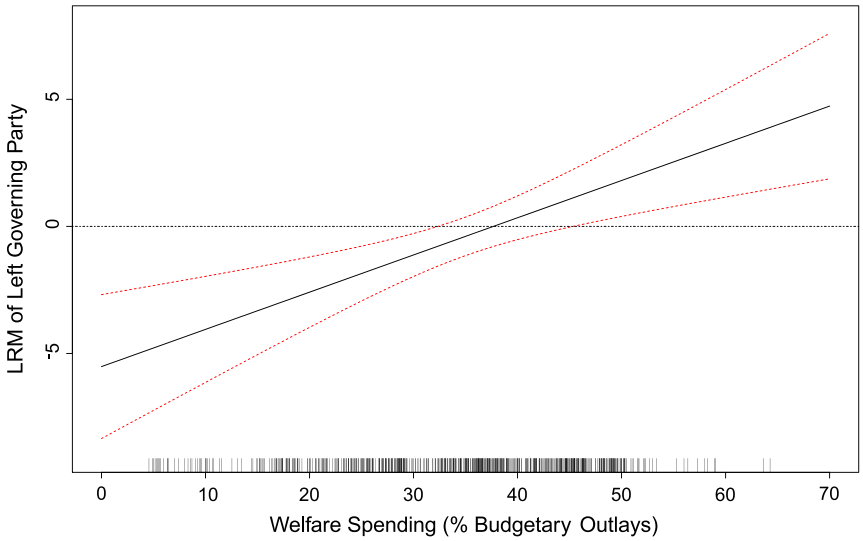

Figure 2 Partisan effect on environmental policy output expansion (based on Model 8). LRM, long-run multiplier.

Model 9, we consider the adoption of a new environmental policy as an independent event. We then model the log odds of the event, a binary variable, as a linear combination of the same set of independent and control variables. In this static logistic regression, all economic indicators, as well as the welfare-spending variable, are included in historic five-year running averages. The coefficient estimate of the interaction term is positive and significant at a 10 percent level. At high levels of welfare spending (> 29 percent of total government outlays), the presence of a left-wing government increases the log-odds of the environmental policy adoption. Again, this is consistent with our main hypotheses.

Finally, we present an additional model that explores the applicability of our theoretical framework to a different dimension of environmental policy output: the stringency. In Model 10, the stringency of a policy (level of environmental tax) is modeled as opposed to the scope of policies. Because of the availability of environmental tax data, we have a short panel (1995–2005). We thus use the generalized method of moments (GMM) estimator (Arellano & Bond 1991), which addresses potential inconsistencies in the fixed-effects estimator in dynamic models with a short time span.[18] Our theoretical narrative about the interaction effect holds. Having a left-wing government is positively associated with environmental tax per capita under large (> 30 percent) welfare spending.

Our finding of the non-uniform partisan effect on environmental tax is also consistent with the existing literature on environmental taxation. On the one hand, an environmental tax can be an attractive instrument for right-wing governments that prefer limited progressivity as such a tax tends to be regressive (Wier *et al.* 2005; Kerkhof *et al.* 2008). On the other hand, an environmental tax, can be attractive to a left-wing government that has already exhausted progressive income taxes as a feasible policy instrument. The left-wing government can impose new regressive taxation while maintaining progressive income tax revenue at the highest politically feasible level (Daugberg & Svendsen 2001; Kato 2003).

5. Conclusion

In this paper, we seek to move beyond the analytical parallels between the welfare state and the environmental state and explore the dynamic interaction between the two. Standard accounts of how partisanship affects environmental policymaking tend to assume that left-wing governments, ceteris paribus, are more likely than their right-wing counterparts to adopt environmental regulations. In contrast, we argue that governments develop their environmental policy commitments by evaluating how the new environmental policy might affect the pursuit of their distributive commitments in the context of the existing welfare state. Accordingly, our theoretical framework posits a non-uniform effect of government partisanship on environmental policy output.

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We test hypotheses derived from the standard account, as well as from our own theoretical model, and find support for the latter. A left-wing government is more likely to increase environmental policy output when it inherits and can maintain generous welfare spending. Large welfare spending mitigates the direct costs of a new environmental policy accrued to labor and/or reduces the opportunity costs of environmental policy expansion perceived by the left-wing government. Under a small welfare state, a right-wing government is more likely to increase environmental policy output. A small welfare state allows more flexibility for businesses in recovering regulatory costs and/or reduces the opportunity costs of environmental policy expansion perceived by the right-wing government.

Our study has two important implications for the study of the environmental state. First, it provides a clear illustration of how the environmental state has emerged as an additional layer on top of the welfare state, rather than as an analogous yet separate entity. In fact, we find that the welfare state plays a pivotal role in translating demands for state action on environmental problems into regulatory responses. The effect of the welfare state, however, does not follow static welfare regime typologies (cf. Koch & Fritz 2014). Rather, it is a dynamic interplay between the welfare state and the incumbent government partisanship that determines the trajectory of environmental policy expansion.

Second, the results of our study raise some larger issues about the capacity of contemporary states for addressing environmental degradation in general. Our findings indicate that a new environmental policy is introduced more easily under certain domestic political conditions than in others, even in the context of the most advanced democracies in the world. To some, this finding might be viewed as to suggest that the state in general and representative democracy in particular cannot be trusted to act consistently as an instigator of far-ranging environmental policy expansion. While such skepticism may be warranted, a better understanding of the political conditions that affect the likelihood of environmental policy expansion can empower various non-state actors (be it domestic or transnational) to more strategically influence the process of environmental policy expansion. As long as environmental demands continue to grow, different configurations of political conditions can open up different paths to environmental state development.

This study also points to areas for future research on the environmental state. While we focus on environmental regulatory expansion as a single analytical dimension of the environmental state, future studies can explore variation in types of regulatory instruments (e.g. first-generation command and control regulations vs. second generation regulations) and also bring in non-regulatory (e.g. distributional) aspects of the environmental state. How would the political economy of the welfare state shape these different dimensions of environmental state? Also, can we observe the emergence of varieties of environmental states that generate distinct complementarities with the existing varieties of welfare states?

Notes

1 While we are making a probabilistic generalization here, we acknowledge that political-ideological posture of unions and workers' movement might vary over time (as industrial structure and patterns of conflict over resources change) and by types of unions (e.g. among skilled vs. unskilled workers and across sectors) (Touraine *et al.* 1987). Such variation might explain the union's stance over a specific environmental policy in a specific political economic context. Also, see case studies of Siegmann (1986, on Germany and the US) and Jahn (1993, on Sweden and Germany) on this matter.

2 The panel is unbalanced because of missing observations in some of our independent variables.

3 Four countries – Mexico, South Korea, Hungary, and Poland – only joined the OECD in the 1990s. Our findings from the main model specification hold when we drop these four countries.

4 The policy output data used in this paper has two sources. The first source is the ENVIPOLCON data set. The original ENVIPOLCON data (Holzinger *et al.* 2008) and its update (Holzinger & Sommerer 2011) have recorded the presence of 21 environmental policies in 24 countries for a complete time series from 1970 to 2005. The second data source is the GRACE data set developed by the GRACE project hosted by the Department of Political Science at Stockholm University. The GRACE data set employs the same policy coding protocol as ENVIPOLCON but adds four new policies, and 13 new countries (primarily emerging economies and non-European countries). We are also in the process of extending the time coverage to 2010. For the purpose of creating a sample of comparable (OECD) cases, non-OECD economies (Romania and Bulgaria) that were included in the original ENVIPOLCON data are excluded from our analysis.

5 Panel unit root test based on augmented Dickey–Fuller regressions suggests that the series is stationary. Proportions are remapped to range from 0.025 to 0.975 in order to include those observations with the score 0 or 25. Those with 0 are all New Zealand observations (1975–1985), and those with 25 are Netherlands in 2003–2005, Hungary in 2005, and France in 2005.

6 Because an independent variable has a linear effect on the logit-transformed DV, its effect in terms of a change in a *raw* DV depends on the initial value of the DV. For instance, the effect of a shift in government partisanship leads to a more substantial expansion of policy portfolio (in raw proportional terms) for countries with mid values of the initial DV (e.g. countries that had

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10–15 out of 25 sample policies), and is attenuated at extremely high initial DV (e.g. countries that already have 23–24 policies in place).

7 We do not have any case of policy repeal in our sample, so country-year-policy area observations posterior to adoption are removed.

8 The data are from the Database of Political Institutions (DPI) by Beck *et al.* (2001).

9 The data are from the International Monetary Fund (IMF). As we plan to extend the coverage of the analysis to emerging economies in a future study, data from the IMF is preferred to others such as the OECD's dataset.

10 GDP per capita data are taken from Haber and Menaldo (2011). Deindustrialization (100-Industry value added as a percentage of GDP) and urbanization (urban population as a percentage of total population) data are from World Bank.

11 Trade (as a percentage of GDP) data are from World Bank. Environmental treaty data are from Mitchell (2002-2016).

12 Unemployment rate data are from World Bank.

13 The data are from the Manifesto Project by Lehmann *et al.* (2016).

14 The ECM specification, as a fully general dynamic model, is algebraically equivalent to autoregressive distributive lag (ADL) specification.

15 Other informative quantities from ECM are the error correction rate, $-\beta 0$, and the mean lags of adjustment. The error correction rate is the rate at which subsequent adjustments (i.e. adjustment following the instantaneous shift) in dependent variable occur. In the case of yearly observations, a high error correction rate would indicate *quick* annual adjustments to the new equilibrium. The mean lags of adjustment tell us how many years it actually takes to complete adjustment. In our case, this may correspond to the average number of years required for a shift in social spending to reshape environmental policy scope. Mean lag length can be calculated as follows:

$$MeanLagLength_x = \frac{\beta_{2x} - \beta_{1x}}{\beta_{2x}} - \frac{\beta_0 + 1}{\beta_0} \tag{3}$$

16 $LRM_{left.government} + (LRM_{Interacton.term} \times Level\ of\ Welfare\ Spending)$

17 This robustness check also excludes the non-Western countries in the sample (Hungary, Mexico, Poland, and South Korea). There were not enough data points on these countries to estimate the correlation coefficient between the left orientation and the welfare orientation.

18 Given the small number observations, the GMM estimator is used on the partial adjustment model (PAM) specification rather than on the full ECM specification to avoid over-fitting. As we do with the ADL specification, we calculate the LRM of each variable:

$$EnvTax_{it} =_{\gamma 0} EnvTax_{i,t-1} + X_{i,t-1\gamma 1} + \alpha_i + \varepsilon_{it} \tag{4}$$

$$LRM_x =_{\gamma 1} / (1 - \gamma 0) \tag{5}$$

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Appendix A

DV	Min	1st Q	Median	Mean	3rd Q	Max
Policy scope raw	0.00	11.00	15.00	14.86	20.00	25.00
Policy scope logit transformed	−3.664	−0.229	0.385	0.452	1.295	3.664
Environmental tax	221	549.2	707	792.1	957	1813
IV						
Left government	0	0	1.0000	0.5745	1	1
Welfare spending	7.021	27.649	36.013	34.730	43.301	64.274
Trade	15.92	46.89	60.42	65.86	74.27	183.62
GDP per capita	6203	16656	20658	20269	24543	36696
Deindustrialization	57.13	66.05	69.31	69.16	72.30	80.12
Urbanization	42.78	67.59	75.65	74.72	82.18	97.29
Environmental administration	0	1	1	0.9652	1	1
Environmental treaty	29	67	92	95.73	120	232
EU accession	0	0	1	0.5648	1	1
Political constraint	0.220	0.510	0.700	0.699	0.880	1.260
Unemployment	0.200	4.610	7.070	7.588	9.590	23.880
Environmental manifesto	0.064	2.510	4.084	4.718	6.315	15.305

Descriptive statistics are based on the observations included in Models 4. Descriptive statistics on Environmental Tax (per capita) are based on the observations included in Model 10. DV, dependent variable; GDP, gross domestic product; IV, independent variable.

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